

Singapore University of Technology and Design
Engineering Systems and Design

ESD 40.012 Manufacturing and Service Operations (MSO)

Summer 2025; Mid-Term Exam date: July 8 2025

Student Name: _____ Student ID: _____

Instructions:

- Two-sided formula sheet
- Scientific calculator (don't program anything)
- Attempt your answers on this booklet (left side ok). Request extra pages if necessary.

Question 1 (25pts):

The monthly sales for Bulesky Inc were as followed:

- a- Forecast January Sales of the following year using: (15pts)
 - a. Naïve method
 - b. A 3-month moving average
 - c. A 6 months weighted average using 0.1,0.1,0.1,0.2,0.2 and 0.3, the heaviest weights applied to the most recent months
 - d. A trend projection based on data from August 2024 to December 2024
- b- With the data given which method would allow you to forecast next March's sales (5pts)
- c- The actual sales in January 2025 was 25. Using MAPE formula identify the most accurate forecast method among a,b,c and d (5pts)

Month	Sales	Forecast
1 January 2024	20	
1 February 2024	21	
1 March 2024	15	
1 April 2024	14	
1 May 2024	13	
1 June 2024	16	
1 July 2024	17	
1 August 2024	18	
1 September 2024	20	
1 October 2024	20	
1 November 2024	21	
1 December 2024	23	
1 January 2025		???

Question 2: 25pts

Bob is a supply planner for Vision Inc, a company which manufactures eye drops. The company has been very successful but competition is fierce though Vision Inc products have a better quality and are recommended in magazines.

Recently there has been a lot of pressure on the cost and everyone is looking at the inventory at the end of each quarter.

Bob is responsible for one of the flagship **product family** and has been told to avoid building any inventory, reject any request of overtime and take serious action to avoid having idling employees. Bob was heavily blamed last quarter as your ending inventory was 1000 units. Bob asked your help.

The demand for the product is as below:

Quarter	Demand
1	5000
2	10000
3	3000
4	6000

Cost elements:

Beginning Inventory: 1,000 units

Hiring Cost per Employee: \$200

Firing Cost per Employee: \$500

Beginning workforce: 60 employees

Inventory Carrying Cost: \$2 per unit per quarter of ending inventory

Stock out Cost: \$5 per unit

Regular Payroll cost: \$1,200 per employee per quarter

Each employee can produce 100 units per quarter

Question 2a: (10 pts)

Based on the strong guidelines you received from Bob what would be the production plan for each quarter.

What will be the associated cost to produce that plan ?

Please fill you response in provided table in the answer sheets.

Period	0	1	2	3	4
Demand		5000	10000	3000	6000
Production Plan					
Ending Inventory	1000				
Backorder	0				
# working employees	60				
# fired employees					
# hired employees					

Cost (period 1 to 4)	
Hiring cost	
Firing cost	
Inventory cost	
Stockout cost	
Payroll cost	
Total Cost	

Question 2b: (10pts)

You believe this is not the best option and you would prefer not to fire anybody but adapt the production accordingly to utilize your resources at best

What would the new plan look like? What would be the associated cost?

Period	0	1	2	3	4
Demand		5000	10000	3000	6000
Production Plan					
Ending Inventory	1000				
Backorder	0				
# working employees	60				
# fired employees					
# hired employees					

Cost (period 1 to 4)	
Hiring cost	
Firing cost	
Inventory cost	
Stockout cost	
Payroll cost	
Total Cost	

Question 2c (5pts)

- 1- Prepare your recommendation for Bob with 3 clear arguments.
- 2- Do you think he will agree to your proposal ? Why or why not ? What action could you trigger in order to minimize risks and cost in the future?

Question 3: 25pts

The company Joo Chiat Inc has been assigned a construction project for the new memorial garden. It has identified 7 major activities to complete the project, labelled from A to G. Some activities need to precede others and the completion time for each is given in below table.

Activity	Completion Time	Predecessor
A	2	-
B	3	-
C	5	A
D	9	B
E	5	C, D
F	3	D
G	2	E

To prepare the work you were asked to map the project length and critical path:

- a) (8pts) Draw the project network (activity on node)
- b) (7pts) Determine the expected project length in weeks

- c) (6pts) Find the critical path
- d) (4pts) In (b) what activity (or activities) have slack? What could be done to reduce the project length?

Question 4: 25pts

Sky Printing has 6 jobs awaiting processing. Processing time and due date are given in the table. The jobs arrived in alphabetical order. Start day is 1.

Job	Processing time (days)	Due date (Days)
A	6	22
B	12	14
C	14	30
D	2	18
E	10	25
F	4	34

Question 4a: 7pts

Set the processing sequence and completion time according to FCFS rule. Identify tardy jobs. Present your answers in tables.

Evaluate the results according to:

- Mean Flow time
- Average tardiness
- Utilization

Question 4b : 7pts

Set the processing sequence and completion time according to SPT rule. Identify tardy jobs. Present your answers in tables.

Evaluate the results according to:

- Mean Flow time
- Average tardiness
- Utilization

Question 4c:7pts

Set the processing sequence and completion time according to CR rule. Identify tardy jobs. Present your answers in tables.

Evaluate the results according to:

- Mean Flow time
- Average tardiness
- Utilization

Question 4d:4pts

Which approach would you use to minimize the number of tardy jobs?

ANSWERS Sheet

Question 1:

Question 2:

Question 2a:

Period	0	1	2	3	4
Demand		5000	10000	3000	6000
Production Plan					
Ending Inventory	1000				
Backorder	0				
# working employees	60				
# fired employees					
# hired employees					

Cost (period 1 to 4)	
Hiring cost	
Firing cost	
Inventory cost	
Stockout cost	
Payroll cost	
Total Cost	

Question 2b:

Period	0	1	2	3	4
Demand		5000	10000	3000	6000
Production Plan					
Ending Inventory	1000				
Backorder	0				
# working employees	60				
# fired employees					
# hired employees					

Cost (period 1 to 4)	
Hiring cost	
Firing cost	
Inventory cost	
Stockout cost	
Payroll cost	
Total Cost	

Question 2c:

Question 3:

Question 4: